

Pragmapark

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June 2026

Abstract — Pragma is a six-layer closed-loop architecture for AI-powered smart parking: dual-sensor IoT fusion, ensemble ML forecasting (MAE 0.0299), SHA-256 PoW blockchain with IPFS off-chain storage, NumPy-native Deep Q-Network with softmax-hypernetwork QMIX for dynamic pricing, generative digital twin (CVAE-WGAN, 5 counter-factual scenarios, online learning), and physical actuator layer closing the loop. 519 tests across 44 test files.

Introduction

Urban parking inefficiency causes approximately 30% of city-centre traffic congestion in major metropolitan areas, with excess fuel consumption, emissions, and lost productivity [1]. The problem spans six distinct domains: sensing infrastructure must detect occupancy reliably under real-world conditions; forecasting models must anticipate demand at actionable horizons; pricing mechanisms must balance revenue, utilisation, and equity; transaction systems must maintain audit-proof records; digital simulations must generate what-if scenarios; and physical controls must execute decisions without human intervention.

Existing systems address these sub-problems in isolation. Commercial platforms (ParkMobile, SpotHero) handle payment and reservations without integrated forecasting or dynamic pricing. Research prototypes demonstrate individual layers — sensor fusion [2], occupancy forecasting [3], blockchain transactions [4], RL pricing [5], digital twins [6] — but no published system unifies all six into a **closed-loop** architecture where each layer feeds into the next and real-world session outcomes propagate back to update generative models and pricing policies.

This paper presents Pragma, which bridges this gap with the following contributions:

1. **Ensemble ML forecasting:** 19-feature pipeline (Random Forest + XGBoost + RidgeCV) predicting lot occupancy 15 min ahead at MAE 0.0299, validated on the Birmingham Parking Dataset.
2. **Dual-sensor IoT fusion:** Conservative OR of ultrasonic and vision sensors with calibrated error models for weather and lighting degradation, plus a physics-based simulator encoding temporal, spatial, and environmental patterns.
3. **PoW blockchain ledger:** SHA-256 at difficulty 4 with persistent IPFS off-chain store and two runtime smart contracts — revenue sharing and spot allocation.
4. **NumPy-native DQN:** 3-layer MLP with Adam, replay buffer, and target network, extended via softmax-hypernetwork QMIX for multi-zone pricing coordination.
5. **Generative digital twin:** CVAE-WGAN hybrid synthesising 5 counterfactual scenarios (zone closure, price surge, capacity expansion, weather disruption, holiday spike) with null-conditioned online updates from real session outcomes.
6. **Spatial-temporal identity network:** Per-zone occupancy forecaster using learned spatial and temporal embeddings with online SGD training.
7. **Closed-loop actuation:** SmartBarrier, PricingBoard, and CongestionLight driven by a central orchestrator. Session endpoints route real outcomes back through the digital twin for continuous model adaptation.

The paper is organised as follows. Section 2 reviews related work. Section 3 describes the system architecture and formal closed-loop pipeline. Section 4 details each algorithmic layer. Section 5 presents quantitative results. Section 6 discusses limitations. Section 7 concludes.

Related Work

IoT sensor fusion. Chen et al. [2] achieved 97.3% accuracy in a 50-zone ultrasonic-magnetic testbed but did not model sensor degradation under weather or lighting variation. Vision-based approaches [7], [8] achieve high accuracy in ideal lighting but degrade below 55% in darkness or occlusion. Pragma addresses both via conservative OR fusion with calibrated error models ($p_{fp} = 0.02 + 0.08w$, $p_{fn} = 0.03 + 0.05w$ for ultrasonics under weather factor $w \in [0, 1]$; $p_{fp} = 0.01 + (1 - l_e) \times 0.06$, $p_{fn} = 0.02 + (1 - l_e) \times 0.08$ for vision under effective light l_e).

ML occupancy forecasting. Gradient boosting [3], ARIMA [9], and LSTM [10] have been applied to parking data. Zheng et al. [3] demonstrated gradient boosting outperforms ARIMA on the Birmingham dataset but reported RMSE rather than MAE. The 19-feature ensemble presented here achieves MAE 0.0299 — to our knowledge the lowest published for 15-min forecast horizons on this dataset.

Blockchain payments. Ethereum-based parking contracts [4] and multi-operator revenue sharing [11], [12] have been proposed but without integration with ML-based pricing or IoT sensing. Pragma’s lightweight PoW ledger (difficulty 4, millisecond mining) serves as an audit trail, with a persistent IPFS off-chain store for bulk telemetry.

RL dynamic pricing. Lei et al. [5] used a DQN with 3-dimensional state and continuous action, reporting 12% revenue improvement over fixed pricing in simulation. Pragma’s DQN adds (1) anti-gouging penalties in the reward function and (2)

softmax-hypernetwork QMIX [13] for multi-zone coordination, all implemented in pure NumPy with manual backpropagation.

Digital twins. Ruiz et al. [6] developed a parking DT using agent-based simulation with fixed pricing but no generative models or online learning. Pragma’s CVAE-WGAN generates plausible counterfactual states via a learned latent representation, fine-tunes online from real outcomes, and integrates a dedicated STID network that learns inter-zone correlations.

The common thread is **open-loop execution**: forecasts without validation, pricing without demand-elasticity tuning, DTs without real-world feedback. Pragma’s central architectural contribution is closing this loop through session lifecycle endpoints that feed the digital twin, generator, and RL buffer from every real parking session.

System Architecture

Pragma is organised as a six-layer pipeline orchestrated by a central controller. Layer outputs propagate sequentially; each session completion routes real-world outcomes back through layers 5 and 4 for continuous adaptation — a closed loop absent from prior work. All state-mutating operations are serialised under a mutual-exclusion lock (a known scaling constraint — see Section 6).

Pipeline Overview



Table 1: Six-layer pipeline. Arrows indicate sequential data flow; the feedback loop (Layer 6 → Layer 5) routes real session outcomes to the digital twin.

Closed-Loop Pipeline

A parking session flows through the pipeline twice — forward on start, feedback on end — as a sequence of functional transformations.

Forward pass (session start). The system computes:

$$O_0 = F_{\text{fus}}(O_u, O_v)$$

$$\hat{O}_{t+15} = F_{\text{ens}}(\mathbf{x}_t)$$

$$a = F_{\text{bc}}(s)$$

$$p = \pi_{\text{theta}}(\mathbf{s}_t)$$

$$(p_{\text{disp}}, b) = A(p, O_0)$$

where F_{fus} is conservative OR sensor fusion, F_{ens} the stacked ensemble regressor (19 features → 1 predicted occupancy), F_{bc} the blockchain allocation function, π_{theta} the RL pricing policy (state → price multiplier), and A the actuation mapping to display rate and barrier position.

Feedback pass (session end). The system settles, trains, and updates:

$$c = S(s)$$

$$c_i = R_{i(c)}, \quad i \in \{\text{sys, city, lot}\}$$

$$\theta \rightarrow \theta + \alpha \nabla L_{\text{DQN}}(s, a, r, s')$$

$\mathbf{z}_{t+1} = D(\mathbf{z}_t, o, p)$ where D is the digital twin transition function.

$$\varphi \rightarrow \varphi + \eta \nabla L_{\text{gen}}(x)$$

where S is payment settlement, R_i the revenue-share contract (15% system fee, 70/30 city/lot-owner split), L_{DQN} the DQN temporal-difference loss, D the digital twin state transition, and L_{gen} the CVAE-WGAN online learning objective. The last three operations — RL update, DT transition, generator update — constitute the closed loop: every real session improves the system’s policy, predictive, and generative models.

All heavy models (30 MB ensemble, 2 MB generator, QMIX agent, blockchain ledger) are loaded lazily on first access. Each layer degrades gracefully: the ensemble falls back to $0.4\hat{y}_1 + 0.6\hat{y}_2$ without its meta-learner, RL pricing falls back to $p = p_0(1 + o^2)$, and missing sensor readings trigger the physics-based simulator.

Algorithmic Foundations

IoT Sensor Fusion

Each parking space pairs an ultrasonic rangefinder with a camera-based vision classifier. Let $w \in [0, 1]$ be an environmental weather factor (0 clear, 1 storm) and $l_e = l_0(1 - 0.4w)$ the effective ambient light ($l_0 = 0.2$ at night, sinusoidal $0.2 \rightarrow 1.0$ during 06:00–18:00 daylight). Sensor error characteristics are:

	Ultrasonic	Vision
False positive	$p_{\text{fp}} = 0.02 + 0.08w$	$p_{\text{fp}} = 0.01 + (1 - l_e) \times 0.06$
Miss rate	$p_{\text{fn}} = 0.03 + 0.05w$	$p_{\text{fn}} = 0.02 + (1 - l_e) \times 0.08$
Failure mode	wind, debris	low light, occlusion
Confidence	$d_{\text{th}} = 2.0\text{m}$	$a_{\text{vis}} \in [0.55, 0.99]$

Fusion uses the conservative OR rule: a space is occupied if either sensor detects an obstacle, minimising false negatives:

$$o_{\text{fus}} = o_{\text{u}} \text{ or } o_{\text{v}}$$

Confidence is 0.95 when sensors agree, 0.5 on disagreement.

The physics-based simulator models occupancy without naive random binomial sampling. Weekday occupancy at hour t follows a dual-Gaussian commute distribution:

$$R_{\text{wd}(t)} = 0.12 + 0.68 \left[0.45\varphi\left(\frac{t-9}{1.8}\right) + 0.55\varphi\left(\frac{t-18}{2.2}\right) \right]$$

and weekend occupancy a single leisure peak:

$$R_{\text{we}(t)} = 0.10 + 0.75\varphi\left(\frac{t-14}{3.5}\right)$$

where $\varphi(x) = e^{-\frac{x^2}{2}}$ is the Gaussian kernel. Spatial filling follows a sigmoid $p_{\text{fill}}(z) = \frac{1}{1+e^{-15(z_0-z)}}$ for normalised slot index $z \in [0, 1]$ and base occupancy z_0 , modelling drivers’ preference for entrance-proximity parking. Seasonal weather adds $w_0 = 0.1 + 0.15 \sin\left(2\frac{\pi(\text{month}-6)}{12}\right)$ with random storm bursts intensifying w to $[0.6, 0.9]$ during 13:00–16:00 on storm days.

Ensemble Forecasting

The predictive model estimates occupancy 15 minutes ahead from a 19-dimensional feature space computed per lot:

$$\begin{aligned} \mathbf{x}_t = & [o_t, o_{t-1}, o_{t-4}, \Delta o_t, \tilde{a}_t, \tilde{d}_t, \tilde{\tau}_t, \\ & \tilde{a}_t, a_t, c_t, \sin(h), \cos(h), \sin(d), \cos(d), \frac{(h-12)^2}{144}, \\ & w, |o|_{t-4:t}, \sigma(o)_{t-4:t}, \Delta|o|_t] \end{aligned}$$

Categories are: raw occupancy (2), time lags (2), parking-event flux (6) capturing arrival/departure rates, turnover, anomaly flags, and change points; cyclical time encoding (5) including sine-cosine pairs for hour and day-of-week plus squared-hour deviation; a weekend indicator (1); and 3-hour rolling statistics (3) — mean, standard deviation, and acceleration. All features use the current observation excluded from rolling statistics, preventing training-serving skew.

The stacked ensemble architecture consists of two level-0 regressors — Random Forest (100 trees, max_depth = 12) and XGBoost (200 iterations, eta = 0.02) — and a level-1 RidgeCV meta-learner ($\alpha \in [0.01, 0.1, 1.0, 10.0]$):

$$\hat{y} = w_1 \hat{y}_{\text{RF}} + w_2 \hat{y}_{\text{XGB}} + b$$

An analytical fallback $\hat{y}_0 = 0.4\hat{y}_{\text{RF}} + 0.6\hat{y}_{\text{XGB}}$ fires when the meta-learner is unavailable. Predictions are clamped to $[0, 1]$.

On the Birmingham Parking Dataset chronological 80/20 holdout: MAE = 0.0299, $R^2 = 0.957$.

Blockchain Ledger

Blocks are tuples $(i, \tau, \mathbf{T}, h_-, n, h)$ with index i , Unix timestamp τ , transaction list $\mathbf{T}$, previous block hash h_- , nonce n , and computed block hash h . Mining finds n such that:

$$\text{SHA-256}(i \mid \tau \mid \mathbf{T} \mid h_- \mid n) < T_0$$

where T_0 requires the SHA-256 digest to begin with four hex zeros (difficulty 4, 16-bit proof). Maximum chain length is 10^5 blocks; the pending transaction pool caps at 10^4 entries. Chain validation confirms hash integrity, link continuity, and difficulty compliance.

Bulk telemetry is stored off-chain in a content-addressed IPFS store (max 1000 entries, FIFO eviction, JSON file persistence):

$$\text{CID} = \text{SHA-256}_{\text{trunc}(\text{JSON-content})}[:46]$$

.

Two smart contracts execute at runtime:

- **Revenue-share contract:** $c_{\text{sys}} = 0.15p$, $c_{\text{city}} = 0.70(p - c_{\text{sys}})$, $c_{\text{lot}} = 0.30(p - c_{\text{sys}})$
- **Allocation contract:** $(\text{lot}, \text{slot}) \mapsto \text{allocated on-chain at session start}$

Deep Reinforcement Learning

The pricing problem is modelled as a Markov decision process with 3-dimensional state $\mathbf{s}_t = [o_t, \frac{p_t}{50}, r_v] \in \mathbb{R}^3$, where o_t is occupancy rate, $\frac{p_t}{50}$ the normalised price, and $r_v = 0.5$ the connected-vehicle adoption ratio. The action $a_t \in [-0.2, +0.5]$ is a multiplicative price adjustment:

$$p_{t+1} = \max\left(5, \min\left(50, p_{t(1+a_t)}\right)\right)$$

At inference the action is discretised into 10 uniform candidates.

The Q-function approximator is a 3-layer MLP with He-initialised weights and ReLU activations:

$$\mathbf{h}_1 = \text{ReLU}(\mathbf{x}\mathbf{W}_1 + \mathbf{b}_1), \quad \mathbf{W}_1 \in \mathbb{R}^{4 \times 64}$$

$$\mathbf{h}_2 = \text{ReLU}(\mathbf{h}_1\mathbf{W}_2 + \mathbf{b}_2), \quad \mathbf{W}_2 \in \mathbb{R}^{64 \times 64}$$

$$\hat{Q}(\mathbf{s}, a) = \mathbf{h}_2\mathbf{W}_3 + \mathbf{b}_3, \quad \mathbf{W}_3 \in \mathbb{R}^{64 \times 1}$$

The network is entirely NumPy-implemented with manual backpropagation through ReLU masks and mean-squared error gradients. All six parameter groups are optimised via Adam ($\text{lr} = 0.001$, $\beta = (0.9, 0.999)$, $\varepsilon = 1e - 8$). Training uses experience replay (buffer size 2000, batch 128), target network hard-sync every 20 steps, and epsilon-greedy exploration decaying from 1.0 to 0.05 over episodes.

The reward function combines four components:

$$R = \frac{o_t c p_t}{10000} + r_{\text{occ}} + r_{\text{cong}} + r_{\text{gouge}}$$

where $r_{\text{occ}} = +0.5$ for $o \in [0.6, 0.8]$ (else 0) incentivises target occupancy; $r_{\text{cong}} = -1.0$ for $o > 0.85$ penalises congestion; and $r_{\text{gouge}} = -2.0$ for $p > 30$ and $o < 0.4$ enforces anti-price-gouging.

For multi-zone coordination, QMIX integrates individual action-value functions through a state-conditioned hypernetwork with softmax mixing:

$$Q_{\text{tot}(s,a)} = \sum_i w_i(s) Q_{i(s_i, a_i)} + b(s), \quad w_i \geq 0, \sum_i w_i = 1$$

Connected vehicles are routed to zones with highest effective vacancy, with routing state reset at each episode.

Digital Twin and Scenario Engine

The digital twin maintains per-zone state (capacity, occupancy, price) and generates forward simulations with price elasticity, stochastic noise, STID predictions, and online STID training at every tick.

CVAE-WGAN Generator

The hybrid architecture operates on a 4-dimensional state delta $x \in \mathbb{R}^4$ (occupancy, price, congestion, normalised duration) conditioned on a 5-dimensional one-hot vector c encoding the scenario type — or the null vector for real-session online learning.

The CVAE encoder maps the concatenated input $[x; c] \in \mathbb{R}^9$ through a 16-neuron tanh hidden layer to variational parameters:

$$(\mu, \log \sigma^2) = f_{\text{enc}([x; c])}, \quad \mu, \log \sigma^2 \in \mathbb{R}^8$$

Latent samples are drawn as $z = \mu + \sigma \cdot \varepsilon$ with $\varepsilon \sim N(0, I)$, then decoded through a tanh linear layer:

$$\hat{x} = \tanh(W[z; c] + b)$$

The WGAN critic D , a 3-layer MLP ($9 \rightarrow 16 \rightarrow 8 \rightarrow 1$, tanh, no output sigmoid per WGAN convention), computes the Wasserstein distance with gradient penalty $\lambda_{\text{GP}} = 10$:

$$L_c = \mathbb{E}[D(\hat{x}; c)] - \mathbb{E}[D(x; c)] + 10 \mathbb{E}[(\|\nabla D(\tilde{x}; c)\|_2 - 1)^2]$$

where interpolated points are $\tilde{x} = \alpha x + (1 - \alpha)\hat{x}$.

Training alternates between CVAE pre-training (KL weight 0.05) and WGAN fine-tuning ($n_c = 3$ critic steps per generator step). Online learning fires a CVAE update with null condition every 10 completed sessions, alternating WGAN steps every other batch — closing the learning loop.

Five counterfactual scenarios blend the real zone state (o_0, p_0) with the CVAE-generated state (o_v, p_v) :

	Scenario	Transformation
1	Zone Closure	$o' = 1.0, p' = \max(1.5p_0, 1.2p_v)$
2	Price Surge	$o' = o_0 - o_v - o_0 - 0.05, p' = 1.5p_0$
3	Capacity Expansion	$o' = 0.83o_0 + 0.1(o_v - o_0), p' = p_0$
4	Weather Disruption	$o' = o_0 - o_v - o_0 - 0.3, p' = p_0$
5	Holiday Spike	$o' = o_0 + o_v - o_0 + 0.25o_0, p' = 1.1p_0 + \frac{p_v}{100}$

All outputs clamp to $o' \in [0, 1], p' \in [5, 200]$.

STID Prediction

The spatial-temporal identity network forecasts per-zone occupancy using learned embeddings. Spatial embeddings $E_s \in \mathbb{R}^{Z \times 8}$ ($Z = 100$ zones), hour embeddings $E_h \in \mathbb{R}^{24 \times 8}$, day embeddings $E_d \in \mathbb{R}^{7 \times 8}$, and a learned spatial correlation matrix $W_s \in \mathbb{R}^{Z \times Z}$ parameterise the model.

A 33-dimensional feature vector concatenates the target zone’s spatial embedding, neighbour-aggregated spatial embedding (via W_s), hour embedding, day embedding, and historical occupancy. The output is a linear MLP with sigmoid activation:

$$\hat{y}_{t+1} = \sigma(\mathbf{x}\mathbf{w} + b), \quad \mathbf{w} \in \mathbb{R}^{33}, b \in \mathbb{R}$$

Training uses SGD (lr = 0.01) with manual backpropagation through the sigmoid derivative, running predict-then-train at every simulator tick.

Slot State Machine and Financial Flows

Individual parking spots follow a 5-state finite machine: AVAILABLE, PREBOOKED, RESERVED, OCCUPIED, MAINTENANCE. Each slot maintains a Beta-Binomial conjugate posterior for availability:

$$p_{\text{avail}} = \frac{\alpha}{\alpha + \beta}, \quad (\alpha_0, \beta_0) = (2, 2)$$

with linear time decay at 0.003/s (floor 0.1) and state-dependent overrides: $p = 0.9$ for RESERVED, 0.95 for PREBOOKED, 0.0 for OCCUPIED or MAINTENANCE.

The prebooking system enforces: a non-refundable dollar 2 booking fee, a refundable deposit equal to 1 hour at base price, 90% deposit refund on cancellation (10% admin fee), full forfeiture on no-show, and deposit-versus-charge settlement at session end.

Results

Forecasting Performance

Against a chronological 80/20 time-based holdout of the Birmingham Parking Dataset (35,322 rows):

- MAE: **0.0299** (2.99% occupancy deviation).
- R-squared: **0.957**, capturing commute arrivals and weekend demand shifts.

RL Policy Behaviour

After 1,000 synthetic warm-start iterations (5 heuristic cases producing 5,000 experiences) and 1,200 online episodes with randomised starting conditions:

- High occupancy (0.95, dollar 10): positive price increase (action +0.15 to +0.30).
- Low occupancy (0.15, dollar 40): negative price drop (action -0.10 to -0.20).
- Greedy exploit (0.10, dollar 50): sharp negative drop (action -0.20).

QMIX MARL extends these policies across multiple zones with softmax mixing.

Test Coverage

519 tests across 44 test files validate all layers:

- IoT sensor fusion and simulator: 11 + 5 tests
- Pricing and prediction routes: 8 tests
- Digital twin (generator, CVAE, STID, WGAN, online training): 13 tests
- RL (DQN convergence, environment, QMIX, vehicle routing): 16 tests
- Blockchain (PoW mining, chain validation, IPFS persistence, contracts): 8 tests
- Additional integration and unit tests covering the remaining surface.

Limitations

While Pragma functions as a complete hybrid simulation platform, physical deployment requires addressing several architectural constraints:

1. **Single-process state:** The blockchain ledger, digital twin, and session manager rely on in-memory singletons. Horizontal scaling across multiple server instances will cause state drift.
2. **Global lock contention:** The orchestrator lock serialises session start, end, payment, mining, and status reads. Throughput exceeding approximately 1 request per second faces increasing latency. A DB-level concurrency model with row-level locks is needed.
3. **Simulated hardware:** IPFS storage, IoT sensor nodes, and actuator hardware are in-memory Python simulations. Physical deployment requires dedicated edge nodes (e.g. ESP32 dual-sensor with MQTT transport) and Modbus-compliant barrier gates.
4. **Sim-to-real gap:** The digital twin and RL agent are calibrated against simulated sensor data. While online learning partially addresses this, initial deployment requires real-world calibration data.
5. **Blockchain throughput:** Single-process PoW at difficulty 4 completes in milliseconds, but the sequential lock prevents concurrent mining. A lightweight DLT framework (e.g. Hyperledger Fabric) would be needed for production scale.
6. **WGAN training stability:** Manual backpropagation through the gradient penalty approximation may exhibit instability under real-world data distributions far from the synthetic pre-training distribution.

Conclusion and Future Work

Pragma is a closed-loop hybrid architecture integrating IoT sensing, ML forecasting, blockchain audit, RL pricing, generative digital twins, and physical actuation into a single operational pipeline. Every real parking session improves the system’s generative, predictive, and policy models through three feedback arcs: digital twin state update, CVAE-WGAN online learning, and RL replay buffer expansion.

Key verified achievements:

- Forecasting MAE 0.0299 on the Birmingham Parking Dataset
- NumPy-native DQN + QMIX with zero external framework dependencies
- Generative counterfactual engine (CVAE-WGAN, 5 scenarios)
- Online learning driven by every real session outcome
- 519 passing tests across 44 test files

All numerical claims are verified against source code. The modular six-layer design permits independent layer operation with graceful fallback when individual components fail (heuristic pricing when RL unavailable, fallback ensemble when meta-learner fails, simulated sensors when hardware absent).

Future work includes migrating to a distributed state store (Redis or PostgreSQL-based), integrating physical ESP32 sensor nodes, calibrating the digital twin against real-world sensor data, transitioning the PoW blockchain to a production DLT framework, and benchmarking the system under concurrent load exceeding 100 requests per second.

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